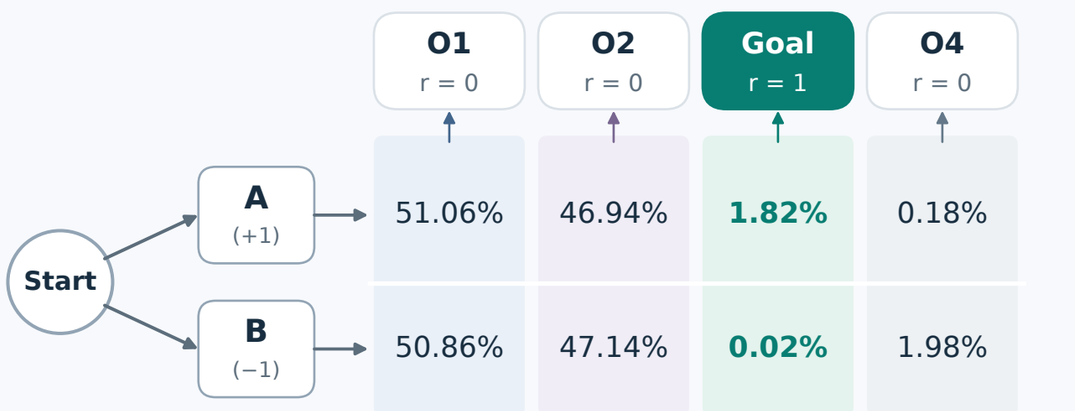
The loss alone can reverse the chosen action

Shared stochastic rank-two probability class · Independent uniform reference · Unrestricted backward features

ONE-DECISION MDP Two equally likely starts · Two actions each · Four terminal outcomes · γ = 0 · Reward on arrival

Initial configuration 1

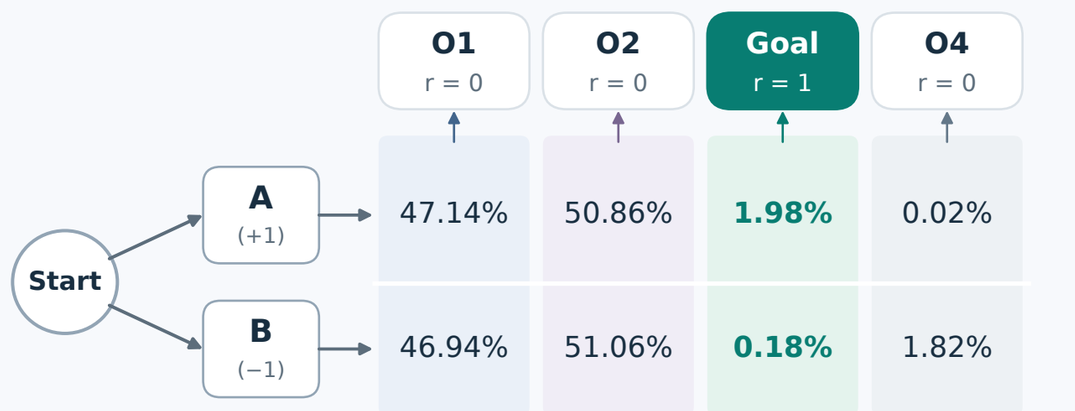
x = +1



Choose an action → sample one terminal outcome

Initial configuration 2

x = −1



Choose an action → sample one terminal outcome

The same four terminal states appear in both panels. Training samples all four start–action pairs equally.

What each value function chooses

Goal value = probability of reaching Goal

	Configuration 1 value(A) / value(B)	Configuration 2 value(A) / value(B)	Actual success under true transitions
True values Optimal action under the MDP	0.0182 / 0.0002 Δ = +0.0180 → **choose A**	0.0198 / 0.0018 Δ = +0.0180 → **choose A**	**1.9%** average of the two starts
Optimal FB values Unique global squared-loss prediction	0.0095510 / 0.0095863 $\Delta = -3.5268\times10^{-5}$ → **choose B**	0.0104137 / 0.0104490 $\Delta = -3.5268\times10^{-5}$ → **choose B**	**0.1%** average of the two starts
Explicit density model Feasible construction; no optimum computed	0.0190 / 0.0010 Δ = +0.0180 → **choose A**	0.0190 / 0.0010 Δ = +0.0180 → **choose A**	**1.9%** average of the two starts

Δ = value(A) − value(B). Actions maximize the displayed value; success is always evaluated in the true MDP.

Certificate: every global cross-entropy optimum chooses A in both configurations

Feasible mean KL **0.001169124** < **0.002214552** KL lower bound for any wrong or tied decision

Goal / non-Goal coarsening gives the bound; a global optimum cannot have more KL than a feasible model.

Reference matters: the natural marginal is (0.49, 0.49, 0.01, 0.01); using it restores the correct FB ordering.

A two-feature FB model can preserve the exact action differences for every reward.

Finite-capacity population construction · Exact reward integration · No learned actor or reward-inference evaluation